INCH-POUND

MIL-R-6106/21J(USAF)
26 April 1993
SUPERSEDING
MIL-R-6106/21H(USAF)
23 March 1990

MILITARY SPECIFICATION SHEET

(J) RELAYS, ELECTROMAGNETIC, TYPE IER, ESTABLISHED RELIABILITY, TRACK MOUNTED, 5 AMPERES, 2PDT, HERMETICALLY SEALED, POLARIZED, TRANSIENT SUPPRESSED COIL

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.

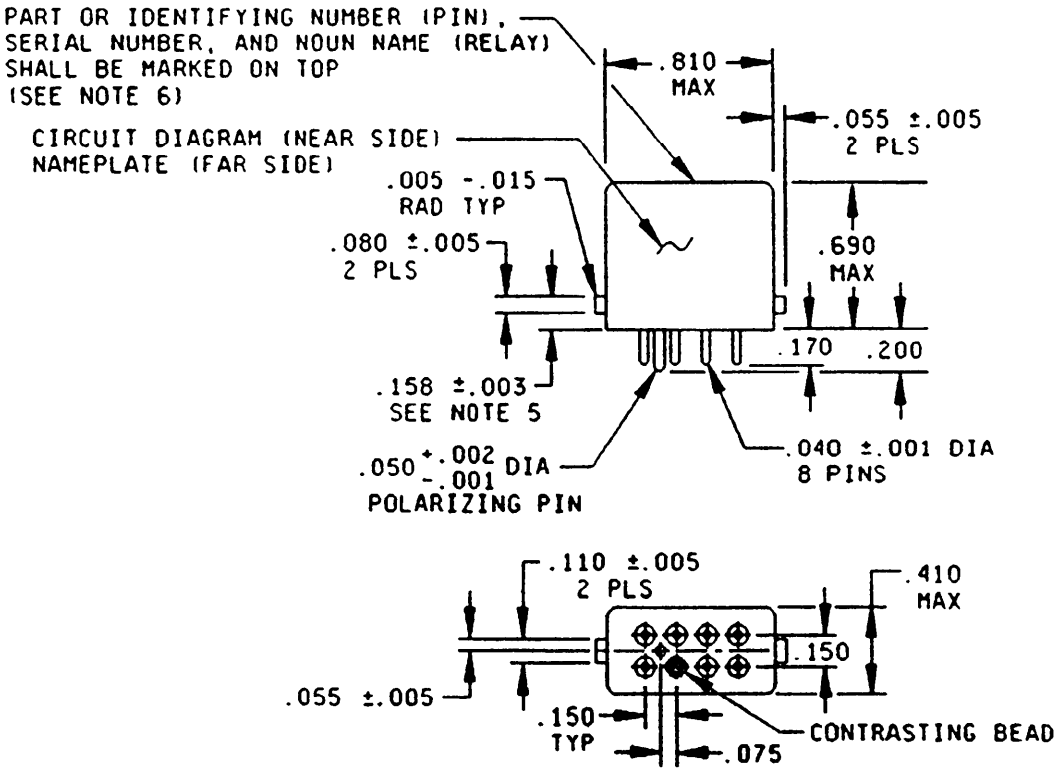
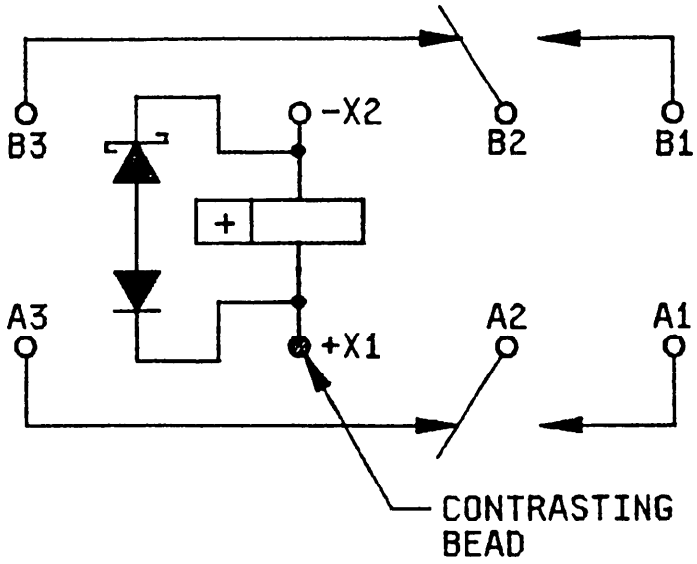


FIGURE 1. Outline drawing.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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CIRCUIT DIAGRAM

When energized with polarity indicated, contacts will transfer

Inches	mm	Inches	mm
.001	0.03	.080	2.03
.002	0.05	.110	2.79
.003	0.08	.150	3.81
.005	0.13	.158	4.01
.015	0.38	.170	4.32
.040	1.02	.200	5.08
.050	1.27	.410	10.41
.055	1.40	.690	17.53
.075	1.91	.810	20.57

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a legible circuit diagram that identifies each terminal location specified.
5. Measure from surface of header.
6. (J) One copy of variables test data shall be supplied with each serialized device. Actual measured values shall be shown for pickup voltage, dropout voltage, contact bounce, operating and release times, contact voltage drop or resistance, coil resistance or current, and electromagnetic interference (dc).

FIGURE 1. Outline drawing - Continued.

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REQUIREMENTS: 1/

COIL DATA:

Nominal operating voltage: 28 V dc.

Maximum operating voltage: 29 V dc.

Pickup voltage:

18.0 V dc maximum over the temperature range.
19.8 V dc maximum during high-temperature test.
22.5 V dc maximum during continuous current test.

Hold voltage: 7.0 V dc. 2/

Dropout voltage: 1.5 V dc. 2/

Duty: Continuous.

Maximum induced transient voltage (back EMF): 42 volts.

Coil resistance at +25°C: 500 ohms ±10 percent.

Coil current at +25°C: 0.065 ampere maximum.

Peak inverse voltage: Diodes shall have a peak inverse voltage of 600 V dc minimum.

Electromagnetic interference: Applicable.

CONTACT DATA:

Contact drop maximum: 150 millivolts initial, 175 millivolts after life test.

Intermediate current: Applicable in accordance with MIL-R-6106.

Overload: 20 amperes dc, 30 amperes ac.

Rupture: 25 amperes dc, 40 amperes ac.

Minimum operating cycles (life):

100,000 at rated resistive, motor, and lamp loads.

20,000 at rated inductive load.

400,000 at 25 percent rated resistive load.

Rated contact load (endurance): See table I.

1/ For qualification only, the relay shall meet the test requirements of this specification when installed in relay socket mount MIL-R-6106/23 and relay bracket mount MIL-R-6106/25.

2/ Over the temperature range.

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(J) TABLE I. Rated contact load (endurance)(amperes per pole) case grounded.

Type of load	28 V dc (amperes)	115 V ac, 400 Hz (amperes)	115/200 V ac, 400 Hz, 3 phase (amperes)
Resistive- - -	5	5	5
Inductive- - -	3	5	5
Motor- - - - -	2	3	3
Lamp - - - - -	1	N/A	N/A

OPERATIONAL DATA:

Operate time: 0.004 second maximum at +25°C. 0.006 second maximum over temperature range.

Release time: 0.004 second maximum at +25°C. 0.006 second maximum over temperature range.

Contact bounce time: 0.001 second maximum.

Release break bounce (NO only): 0.1 millisecond maximum.

PHYSICAL DATA:

Weight: 0.033 pound (15 grams) maximum.

(J) Seal: Hermetic; relays shall be sealed by welding (laser, TIG (tungsten insert gas), or other suitable means as approved by the qualifying activity).

(J) Construction (internal and external): All welded, except that coil magnet wire to coil lead wire shall be soldered.

(J) Terminal strength:
.040 diameter pins: 5 ±0.5 pounds.
.050 diameter pins: 10 ±1.0 pounds.

Finish:

Body: Painted, bright tin plated, or bright nickel plated.

Terminals: Gold plated in accordance with MIL-G-45204, type II, class 1; underplating: nickel, 50 to 150 microinches thick.

Internal visual: Accomplish on a 100 percent basis prior to final encapsulation of the device to detect and eliminate devices with internal defects or foreign particles that could lead to device failure. The inspection equipment shall include optical equipment of at least 10X magnification and visual standards to permit the inspector to make objective decisions as to the acceptability of the device being examined.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Altitude: 300,000 feet.

Vibration scan: Applicable.

Dielectric withstanding voltage: See table II.

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TABLE II. Dielectric withstanding voltage.

Dielectric withstanding voltage	Contacts	Coil
Before life		
Sea level- - - - - - -	1,000 V rms	1,000 V rms
300,000 feet - - - - -	500 V rms	500 V rms
80,000 feet - - - - -	500 V rms	500 V rms
After life		
Sea level- - - - - - -	750 V rms	750 V rms

Insulation resistance:

100 megohms minimum at 500 V dc initial.
50 megohms minimum at 500 V dc after environmental tests.

Vibration (sinusoidal): 0.06 inch double amplitude, 10 to 55 Hz, 10 G's, 55 to 500 Hz, 20 G's, 500 to 3,000 Hz. No contact opening or closing in excess of 10 microseconds.

Vibration (random): Method 214 of MIL-STD-202, test condition I, letter F (0.2 G^2/Hz, 50 to 2,000 Hz).

Duration: 15 minutes each plane.

Measurement: No contact opening or closing in excess of 10 microseconds.

Shock: 50 G's for 11 milliseconds. No contact opening or closing in excess of 10 microseconds.

Acceleration: 15 G's.

Part or Identifying Number (PIN): M6106/21-003 (plus suffix letter as shown in footnote 1/ of table III). PIN example: M6106/21-003M.

TABLE III. Dash number and characteristics.

Part or Identifying Number M6106/21-	Relay type	Coil suppression
003 1/ 2/ 3/ 4/ 5/	I ER	Applicable

1/ Type I ER relays shall be identified as M6106/21, a dash number from table III, and a suffix letter (M, U, X, or P) designating failure rate level. Failure rate level (percent per 10,000 cycles): M = 1%, U = 0.5%, X = 0.3%, and P = 0.1%.
2/ For group A inspection, operational reliability shall be performed at the alternate low level. Relays which have failed to pass group A inspection shall not be reworked.
3/ JAN TX or equivalent screened semiconductors shall be used. Relays with suppression devices shall continue to operate should the suppression circuit be in a failure mode.
4/ Diodes shall have a continuous rating of 1 ampere minimum at +25°C. The zener diode shall have a power rating of 1 watt minimum at +25°C.
5/ Operational reliability shall be run for a total of 7,500 cycles, 2,500 cycles at room temperature, 2,500 cycles at high temperature, and 2,500 at low temperature.

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SUPERSESSION DATA:

Supersession data: See table IV.

TABLE IV. Supersession data.

Superseded PIN: M6106/21-	Replacement PIN: M6106/21-
001	003 1/
002	003 1/

1/ Type I ER relays shall be identified as M6106/21 and a dash number from table III and a suffix letter (M, U, X, or P) designating failure rate level. Failure rate level (percent per 10,000 cycles): M = 1%, U = 0.5%, X = 0.3%, and P = 0.1%.

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F736-03)

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